

Road Accident Severity Predictor



Machine learning project for predicting accident severity from historical road accident data.

Random Forest Classifier

Interactive Streamlit Dashboard

89.37%

TEST ACCURACY

100

TREES

80 / 20

TRAIN / TEST

PROJECT TEAM

RoadSafe — Road Accident Severity Predictor

01 **AYELA AHSAN**

02 **HIBA FATIMA**

03 **AIMA AFZAAL**

04 **INSHAL BUKHARI**

05 **AIZA PARI**

Project Overview

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What RoadSafe AI is designed to do

PROJECT GOAL

Use historical UK road accident records to classify the severity of an accident as Slight, Serious, or Fatal.

Historical Accident Data



Random Forest Classifier



Predicted Accident Severity

TARGET

Accident_Severity → Slight • Serious • Fatal

MODEL

Random Forest Classifier with 100 decision trees

APPLICATION

RoadSafe AI — interactive Streamlit dashboard

Dataset

Historical road accident records used by the final application

13,962

ACCIDENT RECORDS

33

ORIGINAL COLUMNS

6

MODEL FEATURES

3

SEVERITY CLASSES

Target Variable

ACCIDENT_SEVERITY

Slight
Serious
Fatal

Selected Input Features

Speed limit

Day of Week

Weather Conditions

Road Surface Conditions

Light Conditions

Urban or Rural Area

The final model uses only the six features shown above; the remaining dataset columns are not used as model inputs.

Exploratory Data Analysis

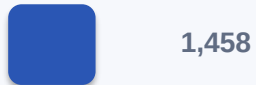
Patterns examined before model training

SEVERITY DISTRIBUTION

Slight



Serious



Fatal

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CLASS IMBALANCE

Slight accidents form the majority of the dataset, so accuracy should be interpreted alongside other classification metrics.

ENVIRONMENTAL CONDITIONS

Weather, light, and road-surface conditions were examined as factors related to accident severity.

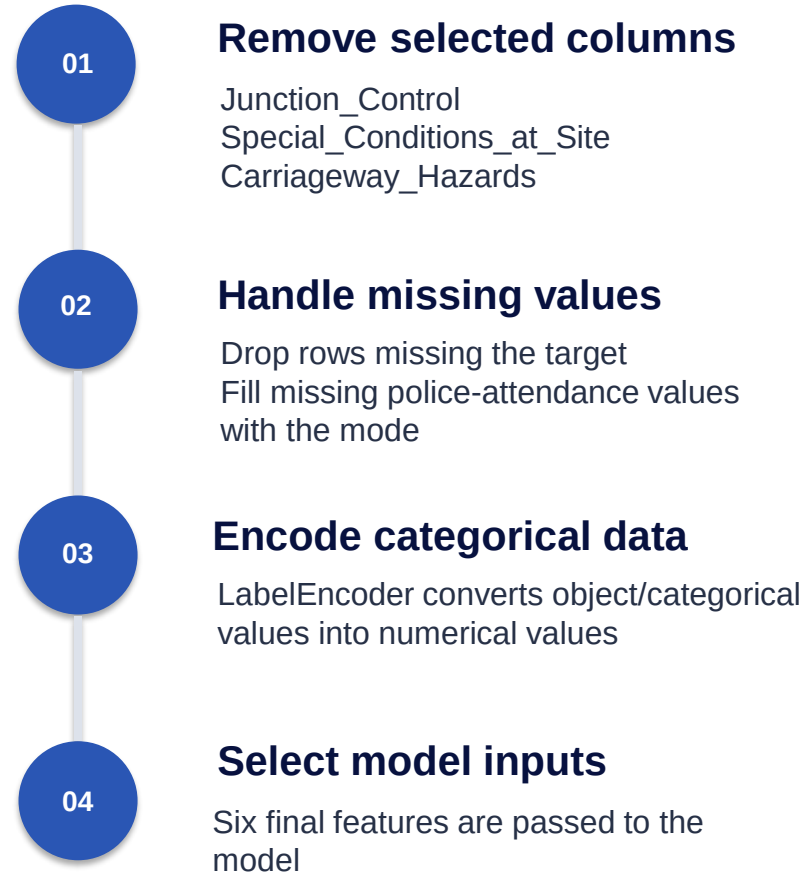
EDA VISUALS

Severity distribution, missing-value analysis, correlation/heatmap analysis, feature importance, and confusion matrix.

Data Preprocessing

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Preparing the historical data for machine learning



Features → Prediction

The six conditions supplied to the final model

Speed limit

Speed limit at the accident location

Day of Week

Day on which the accident occurred

Weather Conditions

Weather conditions at the time

Road Surface Conditions

Condition of the road surface

Light Conditions

Lighting conditions

Urban or Rural Area

Urban or rural area

MODEL OUTPUT

Accident Severity → Slight | Serious | Fatal

Machine Learning Model

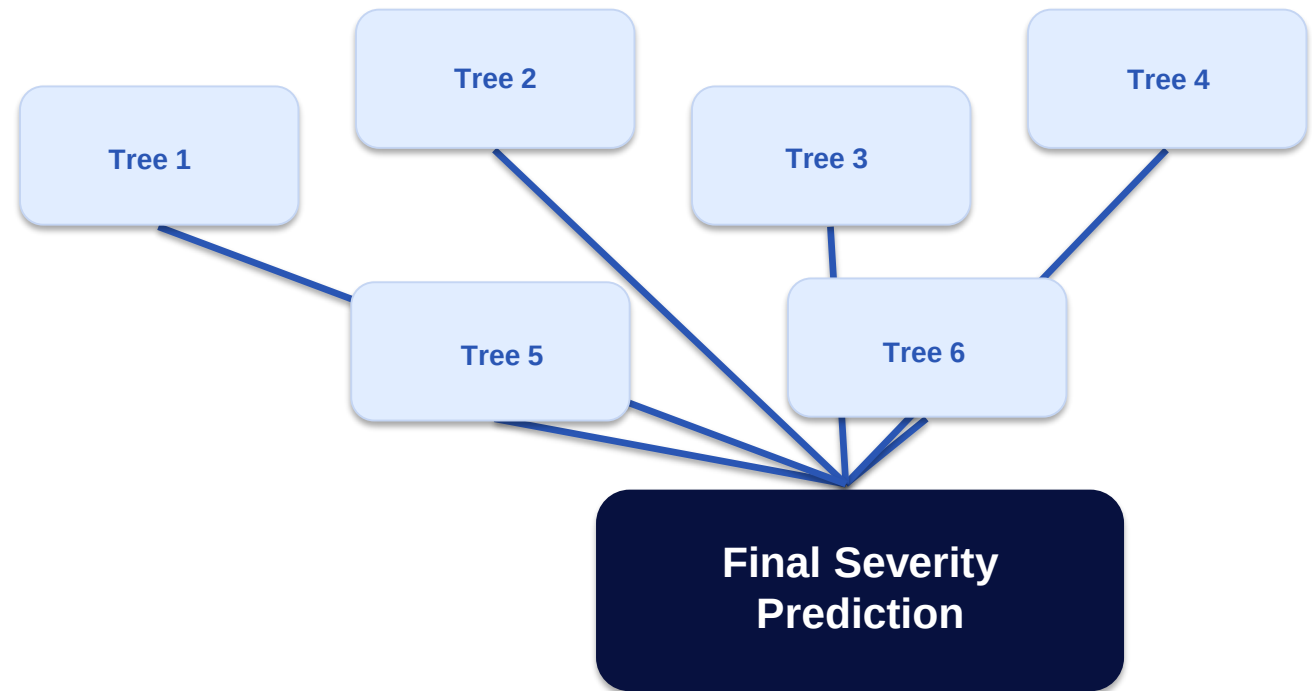
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Random Forest Classifier

MODEL CONFIGURATION

Algorithm	Random Forest Classifier
Trees	100
Random State	42
Train/Test	80% / 20%
Encoding	LabelEncoder

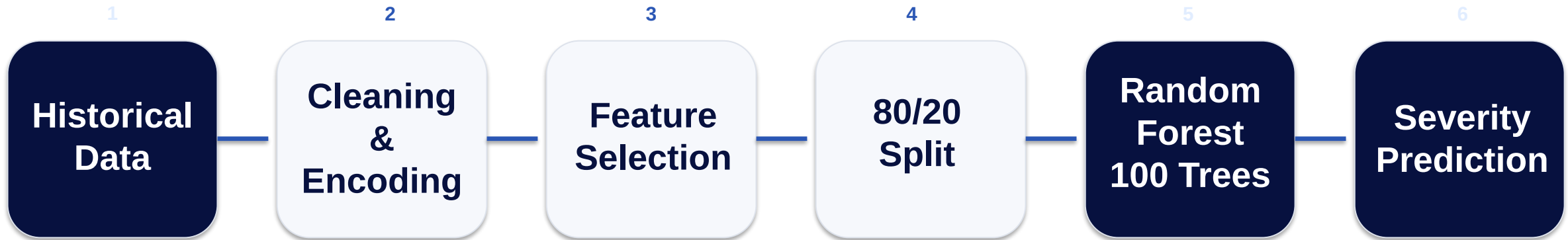
How the Random Forest is used?



100 decision trees are combined to produce the classification.

Machine Learning Workflow

From historical records to a severity prediction



The model learns from the training portion and is evaluated on the unseen test portion.

Model Performance

Evaluation on the 20% test set

89.37%

TEST ACCURACY

11,169

APPROX. TRAIN RECORDS

2,792

APPROX. TEST RECORDS

Evaluation methods

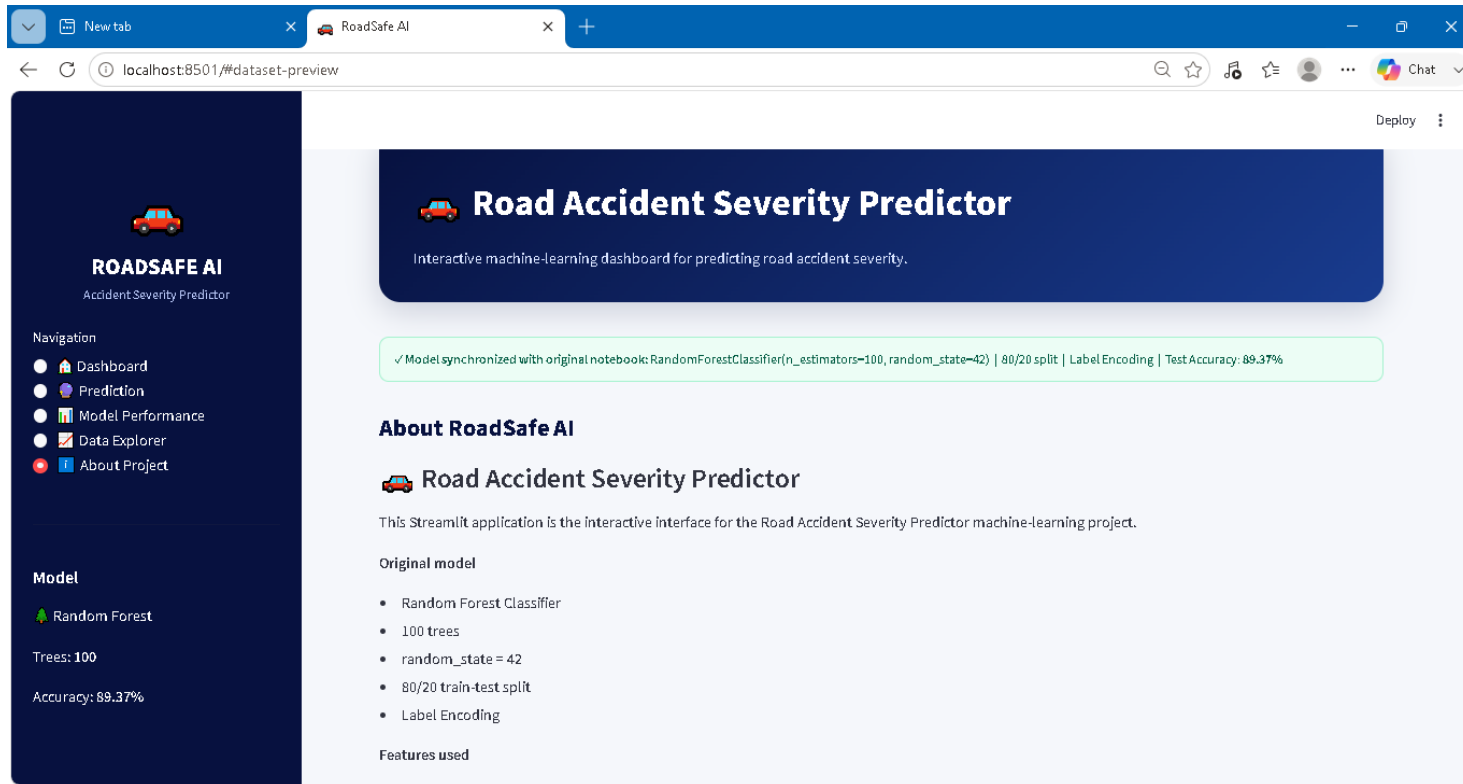
- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

IMPORTANT NOTE

The dataset is imbalanced, so accuracy should be considered together with the classification report and confusion matrix rather than used alone.

RoadSafe Application

Interactive Streamlit interface for the trained model



DASHBOARD

Overview of dataset and model configuration

PREDICTION

Enter accident conditions and get a severity prediction

MODEL PERFORMANCE

View accuracy, classification report and confusion matrix

DATA EXPLORER

Filter and inspect historical accident records

Conclusion

RoadSafe demonstrates an end-to-end machine learning workflow for road accident severity prediction.

DATA

13,962 records
33 columns

MODEL

Random Forest
100 trees

RESULT

89.37%
test accuracy

APP

Streamlit
RoadSafe AI

Future Improvements

- Address class imbalance using SMOTE or class weights
- Experiment with additional models such as Gradient Boosting
- Explore additional data/features and further deployment improvements

THANK YOU!

THANK YOU FOR YOUR TIME
AND ATTENTION.



QUESTIONS & DISCUSSION

